

SimGEA v.1.1.0 manual

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1 Overview

SimGEA is a method for simulating neutral alleles that reflect the population structure observed in an input dataset. The simulated alleles can be analyzed using the same method applied to the empirical data, providing a null distribution that can be used to calibrate p -values.

The workflow consists of two main steps: (i) estimation of a migration matrix from the input allele count data and (ii) simulation of neutral alleles based on the estimated migration matrix. The directories "mig_est" and "null_simu" contain the programs used for these respective steps.

The program in "null_simu" can also run a simple rank-biserial correlation test to assess the association between allele counts and environmental variables. This demonstrates how simulated neutral alleles can be used to obtain calibrated p -values.

SimGEA is not limited to this particular association test. The simulated neutral alleles can be used with other suitable association statistics or analytical methods. The rank-biserial correlation workflow included here is intended as a simple starting point for genotype-environment association analysis using SimGEA.

2 Folder "mig_est" (estimation of a migration matrix)

2.1 Compiling an executable file

The code is written in C++. The Eigen library should be installed in advance. After moving to the folder "mig_est", the executable file can be compiled using the following command:

```
g++ -O3 -std=c++17 main.cpp my_matrix_calc.cpp options.cpp \  
-I /path/to/eigen \  
-o simgea_migration
```

Replace "/path/to/eigen" with the path to the Eigen library on your system.

2.2 Input file

The input file is expected to be a BayPass-style allele count dataset. The file should be a space-separated text file in which each line represents a SNP, and each population is represented by two integers corresponding to the counts of reference and alternative alleles. Missing data is not allowed; therefore, the total number of alleles in each subpopulation must be constant across loci. Sample sizes can differ among subpopulations. Users may apply an imputation method to deal with missing data in preparing the input file.

Example (10 subpopulations, each with 32 alleles):

```

26 6 32 0 32 0 25 7 32 0 32 0 28 4 30 2 31 1 30 2
30 2 31 1 27 5 31 1 27 5 31 1 27 5 32 0 31 1 31 1
14 18 7 25 4 28 3 29 2 30 8 24 12 20 8 24 6 26 2 30
22 10 31 1 29 3 21 11 25 7 31 1 26 6 32 0 29 3 31 1
4 28 5 27 9 23 9 23 1 31 5 27 8 24 6 26 8 24 10 22

```

2.3 Running the program

The compiled program can be run using the following command:

```
./simgea_migration -i <baypass_style_file>
```

The list of options is shown below:

Required:

`-i, --input <STR>` BayPass-style SNP count file (space-separated).

Optional (defaults in [brackets]):

`-o, --out-prefix <STR>` Output prefix [out]
`--maf <DOUBLE>` MAF filter in [0,0.5) [0.0]
`--lr <DOUBLE>` AdamW learning rate [1e-3]
`--beta1 <DOUBLE>` AdamW beta1 [0.9]
`--beta2 <DOUBLE>` AdamW beta2 [0.999]
`--weight-decay <DOUBLE>` AdamW weight decay [0.01]
`--max-step <INT>` Max optimization steps [100000]
`--min-error <DOUBLE>` Relative improvement threshold [1e-6]
`--min-abs-error <DOUBLE>` Absolute improvement threshold [0.01]
`--seed <UNSIGNED INT>` Random seed [randomly determined]
`-l, --log-file <STR>` log file [log.txt]
`-h, --help` Show this help and exit

An important point regarding the minor allele frequency filter is that it is based on the unweighted mean allele frequency across subpopulations, rather than on the pooled allele frequency across all individuals. To clarify this point, let us consider a locus at which a_i out of the b_i sampled alleles are alternative alleles in the i th subpopulation ($i = 1, \dots, n$). In this case, the filter is applied based on $\frac{1}{n} \sum_i \frac{a_i}{b_i}$, rather than on $\frac{\sum_i a_i}{\sum_i b_i}$.

The program searches for a set of pairwise migration rates (or an $n \times n$ symmetric migration matrix, $\mathbf{M}$) that reproduces the covariance matrix estimated from the input data, $\mathbf{\Lambda}_{\text{obs}}$. Let $\mathbf{\Lambda}_{\text{theor}}(\mathbf{M})$ be a covariance matrix calculated from $\mathbf{M}$ using coalescent theory. The distance between these two matrices is evaluated by $D(\mathbf{\Lambda}_{\text{obs}}, \mathbf{\Lambda}_{\text{theor}}(\mathbf{M}))$, where $D(\mathbf{X}, \mathbf{Y}) = \sum_{i < j} (x_{ij} - y_{ij})^2$. The optimization process aims to reduce this distance.

Let $D(t)$ denote the value of the distance at optimization step t . The optimization is terminated when any of the following conditions is satisfied:

- After at least 1000 steps, ten successive steps satisfy $\frac{2D(t-1)-2D(t)}{2D(t-1)+\tau_1} < \tau_1$ and $2D(t-1) - 2D(t) < \frac{2D(0)-2D(1000)}{1000}\tau_2$.
- At $t = 1000k$ ($k \geq 2, k \in \mathbb{N}$), $\frac{2D(t-1000)-2D(t)}{2D(t-1000)+\tau_1} < \tau_1$.
- The total number of optimization steps reaches τ_3 .

Here, τ_1 corresponds to `<min-error>`, τ_2 corresponds to `<min-abs-error>`, and τ_3 corresponds to `<max-step>`.

2.4 Output files

```
"prefix_data_sample_mat.txt":  Covariance matrix of sampled allele frequencies
"prefix_data_pop_mat.txt":    Estimated covariance matrix of population allele frequencies
"prefix_est_mig_mat.txt":     Estimated mig.matrix (2Nm, assuming N haploids)
"prefix_est_cov_mat.txt":     Population covariance matrix calculated from the estimated
                               migration matrix
"prefix_est_sample_mat.txt":   Sample covariance matrix calculated from the estimated migration
                               matrix
```

In the above, prefix is the one specified in the “-out-prefix” option.

3 Folder “null_simu” (simulation of neutral alleles)

3.1 Compiling an executable file

The code is written in C++. After moving to the folder “null_simu”, the executable file can be compiled using the following command:

```
g++ -O3 -std=c++17 main.cpp neutral_simu.cpp gea_func.cpp options.cpp -o simgea_neutral
```

3.2 Input files / parameters

Two input files and one parameter are required. First, a migration file specifies a space-separated $n \times n$ migration matrix that will be used in the simulation. Migration rates should be specified in the unit of N (i.e., $M_{ij} = 2Nm_{ij}$ for N haploids). In the standard SimGEA workflow, use the migration matrix estimated by the “mig_est” program, “prefix_est_mig_mat.txt”.

Second, a BayPass style genotype file is expected. In simulation-only mode (see below), this is used to obtain the total number of alleles in each subpopulation. For this purpose, only the first row of the file is read. In simulation+GEA mode, the program additionally uses the genotype data to calculate p -values for each empirical SNP.

Third, the number of simulated variants per window (or per bin, depending on the mode) must be specified. In the default setting, we assume a frequency-window-based analysis (see below), and this specifies the number of simulated variants belonging to each window.

To run the simulation + GEA mode, the program additionally needs a BayPass-style environmental file. In this file, the environment data are expected to be space-separated, with each row representing one environmental variable.

Example (10 subpopulations, two environmental variables):

```
12.4 15.8 18.1 20.3 22.7 24.5 19.6 16.2 13.9 11.7
8.6 10.2 13.5 16.8 19.1 21.4 17.9 14.7 11.3 9.1
```

The list of options is shown below:

Required:

```
-m, --mig <STR>           Migration file (space-separated).
-d, --data <STR>          Baypass style file (space-separated).
-s, --site <INT>          Total number of variant sites per window
```

Optional (defaults in [brackets]):

```

-o, --output <STR>          Output simulated file [simulated.txt]
--mode <INT>                0: simulation mode, 1: simulation + GEA mode [0]
--sim-mode <INT>            0: window-based, 1 bin-based (allowed in mode 0) [0]
-b --bin-file <STR>         File specifying bin breaks [Required for bin-based mode]
-w, --win-width <DOUBLE>    Width of frequency window [0.025]
--prerun <INT>              Number of trees in the prerun [10000]
--mut-density <DOUBLE>      Expected number of mutations per window per tree [0.1]
--seed <UNSIGNED INT>       Random seed [randomly determined]
-l, --log-file <STR>        Log file [log.txt]
-h, --help                  Show this help and exit

```

Options for +GEA mode (defaults in [brackets]):

```

-e, --env <STR>             Baypass style environment file [Required]
--pval <STR>                Output p-values file [p_vals.txt]
--q-tail <DOUBLE>           The definition of the tail in the generalized Pareto fitting [0.995]

```

3.3 Running the program

This program allows three modes: (i) window-based simulation mode (default), (ii) window-based simulation + GEA mode, and (iii) bin-based simulation mode.

3.3.1 window-based simulation mode (default)

This mode can be run using the following command:

```
./simgea_neutral -m <migration_file> -d <baypass_style_file> -s <site_number>
```

This mode simulates $\lceil S/w_{ac} \rceil$ neutral variants for all possible minor allele counts, where $w_{ac} = \min(2\lfloor xw/2 \rfloor + 1, \lfloor x/2 \rfloor)$. Here, x denotes the total number of sampled haplotypes and S and w are specified by the options `-site` and `-win-width`, respectively. In particular, if $w = 0$, w_{ac} becomes 1, and the program simulates S neutral variants for each possible allele count.

The above method is motivated by the frequency window-based analysis (Figure 1). w_{ac} is the width of window on the allele count scale, and thus each window roughly has a frequency width of $\sim w$. By shifting the window by one allele count at each step, $\lfloor \frac{x}{2} \rfloor - w_{ac} + 1$ overlapping windows are generated. After simulating $\lceil S/w_{ac} \rceil$ variants for each allele count, each window consists of $\sim S$ neutral variants.

In the program, neutral variants are sampled from $\sim S/d$ coalescent trees, where d is specified by the `-mut-density` option. As d decreases, mutations are sampled more sparsely, resulting in lower linkage disequilibrium and longer computation times. In practice, the program first simulates n trees, as specified by `-prerun`, and calculates the total branch lengths corresponding to each minor allele count. Based on this prerun, the mutation rate for each allele count is determined so that approximately S/d coalescent trees are required to simulate $\lceil S/w_{ac} \rceil$ variants.

3.3.2 window-based simulation + GEA mode

This mode runs the GEA analysis following the simulation. A rank-biserial correlation test based on Wendt (1972, *Eur. J. Soc. Psychol.*, 2(4): 463–465) is implemented to test the association between the allelic presence or absence (0/1) and the rank of environmental values.

This mode can be run using the following command:

```
./simgea_neutral -m <migration_file> -d <baypass_style_file> -s <site_number> \
--mode 1 -e <env_file>
```

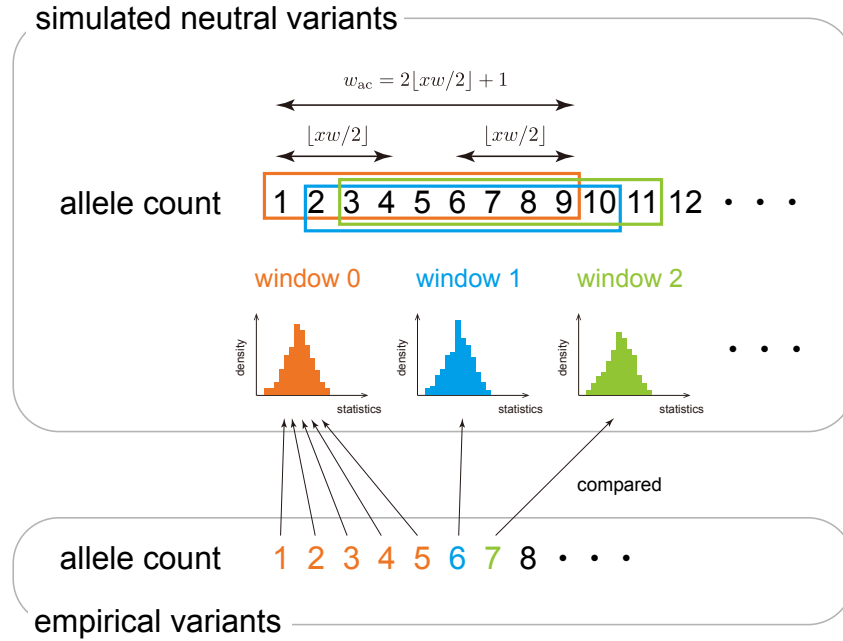


Figure 1: Schematic illustration of the window-based analysis.

Use “-mode 1” to specify this mode.

In the GEA analysis, null distributions of Z values are generated from the simulated variants for each window. The probability density for the bulk is smoothed by the Gaussian kernel, whereas the tail is extrapolated by the generalized Pareto distribution. The definition of the tail can be controlled by the `-q-tail` option. Then, the empirical genotype file is read to calculate p -values. Each empirical SNP is matched to the window with the closest mean minor-allele count.

3.3.3 bin-based simulation mode

The bin-based mode is intended for analyses using non-overlapping minor-allele-frequency bins. This does not support +GEA mode. It is assumed that bins are non-overlapping. In this mode, breakpoints of the bins (minor allele frequency-based) must be provided in the bin file. This file contains one number per line, representing the start of each bin. For example, a bin file

```
0.04999
0.075
0.1
0.125
0.15
0.175
0.2
0.225
0.25
0.275
0.3
0.325
0.35
```

```
0.375
0.4
0.425
0.45
0.475
```

represents 18 bins, consisting of $(0.04999, 0.075]$, $(0.075, 0.1]$, $\dots$, $(0.475, 0.5]$. The program then simulates S neutral variants for each bin. If the bin breakpoints are not ordered correctly or contain invalid bins, the program may update the bin file.

This mode can be run using the following command:

```
./simgea_neutral -m <migration_file> -d <baypass_style_file> -s <site_number> --sim-mode 1 \
  -b <bin_file>
```

Use “sim-mode 1” to specify this mode.

3.4 Output files

```
<output_file>: simulated allele count data
<pval_file>:    list of p-values (only in +GEA mode)
```

The allele count data follow the BayPass-style format, except that the first column represents the minor allele count (window-based mode) or bin index (bin-based mode) (Figure 2). In the pval file, each row represents one variant, and the p -values for different environmental variables are separated by spaces.

total allele count allele counts in allele counts in ...
or bin index the 1st subpop. the 2nd subpop.

54	7	25	2	30	3	29	9	23	9	23	6	26	3	29	4	28	4	28	7	25
19	0	32	4	28	3	29	0	32	5	27	0	32	4	28	3	29	0	32	0	32
132	13	19	21	11	19	13	21	11	20	12	17	15	20	12	17	15	23	9	17	15
132	13	19	21	11	19	13	21	11	20	12	17	15	20	12	17	15	23	9	17	15
127	19	13	10	22	13	19	11	21	12	20	14	18	12	20	15	17	9	23	12	20
49	8	24	2	30	7	25	5	27	5	27	6	26	1	31	7	25	4	28	4	28

Figure 2: The format of the simulated file. The first column represents the minor allele count in window-based mode or the bin index in bin-based mode (starting from 0). The subsequent columns follow the BayPass-style format.

4 Example workflow

In this example, we assume that two input files exist in the current directory

```
"baypass_geno.txt": BayPass-style genotype file
"baypass_env.txt":  BayPass-style environmental file,
```

as well as two executable programs (“simgea_migration” and “simgea_neutral”). The following commands run the entire analysis.

```
# estimation of a migration matrix
./simgea_migration -i baypass_geno.txt
```

```
# simulation-only mode
./simgea_neutral -m out_est_mig_mat.txt -d baypass_geno.txt -s 100000

# simulation + GEA mode
./simgea_neutral -m out_est_mig_mat.txt -d baypass_geno.txt -s 100000 \
  --mode 1 -e baypass_env.txt
```